# Prof. CHOI, Byron Koon Kau

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# About

[Note: choi at hkbu.edu.hk is Dr. Choi's primary work email. Study plans, forms, and requests of letters, etc, should be sent to choi at hkbu.edu.hk.]

Prof. Choi obtained his Ph.D. and MSE degrees in Computer and Information Science from the University of Pennsylvania (UPenn) in 2006 and 2001, respectively. He received his BEng degree in Computer Engineering from the Hong Kong University of Science and Technology (HKUST). Prof. Choi was an Assistant Professor at Nanyang Technological University (NTU), 2005-2008. He was a research associate at the University of Edinburgh in 2005. He visited HKUST theoretical computer science group in 2003. His publications have appeared in IEEE TKDE, VLDBJ, ACM SIGMOD, PVLDB/VLDB, IEEE ICDE, ACM SIGKDD, WWW, ACM CIKM, AAAI, and IJCAI. Prof. Choi was awarded a distinguished program committee (PC) member from ACM SIGMOD 2023 and 2021, respectively, a distinguished reviewer award from PVLDB 2019, an outstanding program committee member for ICDE 2025, and the best reviewers award from ACM CIKM 2021. Prof. Choi received the HKBU President's Award for Outstanding Performance in Individual Teaching in 2022 and the HKBU President's Award for Outstanding Young Researcher in 2016. Prof. Choi was the director of a Croucher Foundation Advanced Study Institute (ASI) titled "Frontiers in Big Data Graph Research" 2015. Prof. Choi has been the ACM SIGMOD information director since September 2021. Prof. Choi is an associate editor of ACM SIGMOD 2027 and PVLDB 2027. He has served on the Editorial Board of Distributed and Parallel Databases (DAPD) as an Associate Editor since March 2025. Prof. Choi is a Fellow of the Hong Kong Institution of Engineers (HKIE) (the Information discipline) a senior member of IEEE, and a professional member of ACM.

# Research Interests

- Graph query processing
- Time series analysis
- Database security
- View update and incremental maintenance

# Selected Publications

- Y. Zhao, L. Xu, J. Jiang, B. Choi, J. Xu and B. He. "ABFlow: Alert Bursting Flow Query in Streaming Temporal Flow Networks." In *Proceedings of the ACM on Management of Data (**SIGMOD '26**) (to appear).*
- L. Xu, J. Jiang, B. Choi, J. Xu and B. He. "Bursting Flow Query on Large Temporal Flow Networks." In *Proceedings of the ACM on Management of Data (**SIGMOD '25**) 18:1-18:26, 2025.*
- R. Zuo, G. Li., R. Cao, B. Choi, J. Xu and S. S. Bhowmick. "DARKER: Efficient Transformer with Data-driven Attention Mechanism for Time Series." In *Proceedings of the VLDB (**PVLDB '24**) 17(11):3229-3242, 2024.*

- J. Jiang, B. Choi, X. Huang, J. Xu and S. S. Bhowmick. "DKWS: An Efficient Distributed System for Keyword Search on Massive Graphs." *IEEE Transactions on Knowledge and Data Engineering (**TKDE**) 39(5):1935-1950, 2024.*
- Y. Peng, B. Choi, T. Chan, J. Yang and J. Xu. "Efficient Approximate Nearest Neighbor Search in Multi-dimensional Databases." In *Proceedings of the ACM on Management of Data (**SIGMOD '23**) 1:(1):54:1-54:27, 2023.*
- L. Xu, B. Choi, Y. Peng, J. Xu and S. S. Bhowmick. "A Framework for Privacy Preserving Localized Graph Pattern Query Processing." In *Proceedings of the ACM on Management of Data (**SIGMOD '23**) 1(2):129:1-239:27, 2023.*
- R. Zou, G. Li, B. Choi, S. S. Bhowmick, D. N-Y. Mah and G. L-H. Wong. "SVP-T: A Shape-level Variable-Position Transformer for Multivariate Time Series Classification." *In Proceedings of the 37th AAAI Conference on Artificial Intelligence* (***AAAI '23***), 11497-11505, 2023.
- G. Li, B. Choi, J. Xu, S. S. Bhowmick, K-P Chun and G. L-H. Wong. "Efficient Shapelet Discovery for Time Series Classification." *IEEE Transactions on Knowledge and Data Engineering (**TKDE**) 34(3):1149-1163,* 2022.
- G. Li, B. Choi, J. Xu, S. S. Bhowmick, K-P Chun and G. L-H. Wong. "ShapeNet: A Shapelet-Neural Network Approach for Multivariate Time Series Classification." *In Proceedings of the 35th AAAI Conference on Artificial Intelligence* (***AAAI '21***), *pages 8375-8383,* 2021.
- P. Yi, B. Choi, S. S. Bhowmick and J. Xu. "AutoG: A Visual Query Autocompletion Framework for Graph Databases." *The VLDB Journal (**VLDBJ**) 26(3):347-372,* 2017.
- Z. Fan, B. Choi, Q. Chen, J. Xu, H. Hu and S. S. Bhowmick. "Structure-Preserving Subgraph Query Services." *IEEE Transactions on Knowledge and Data Engineering (**TKDE**)*, 27(8):2275-2290, 2015.
- D. Wu, B. Choi, J. Xu and C. S. Jensen. "Authentication of Moving Top-k Spatial Keyword Queries." *IEEE Transactions on Knowledge and Data Engineering (**TKDE**)*, 27(4):922-935, 2014.